Native F4/F5, F6 and root solvers

Run 20261019-190236-boolean-adaptive-native-d28a77, source 9bed104b0a. Dedicated c7i.2xlarge, one worker and a reserved complete physical core; four-worker controls reserve two cores/SMT siblings. Exact source/compiler/CPU/ISA and binary manifests remain in the raw tree. Baseline: frozen a78 algorithm with all three previously qualified GF2 switches. Candidate: identity-bound prepared construction and bounded quadratic-tail policies.

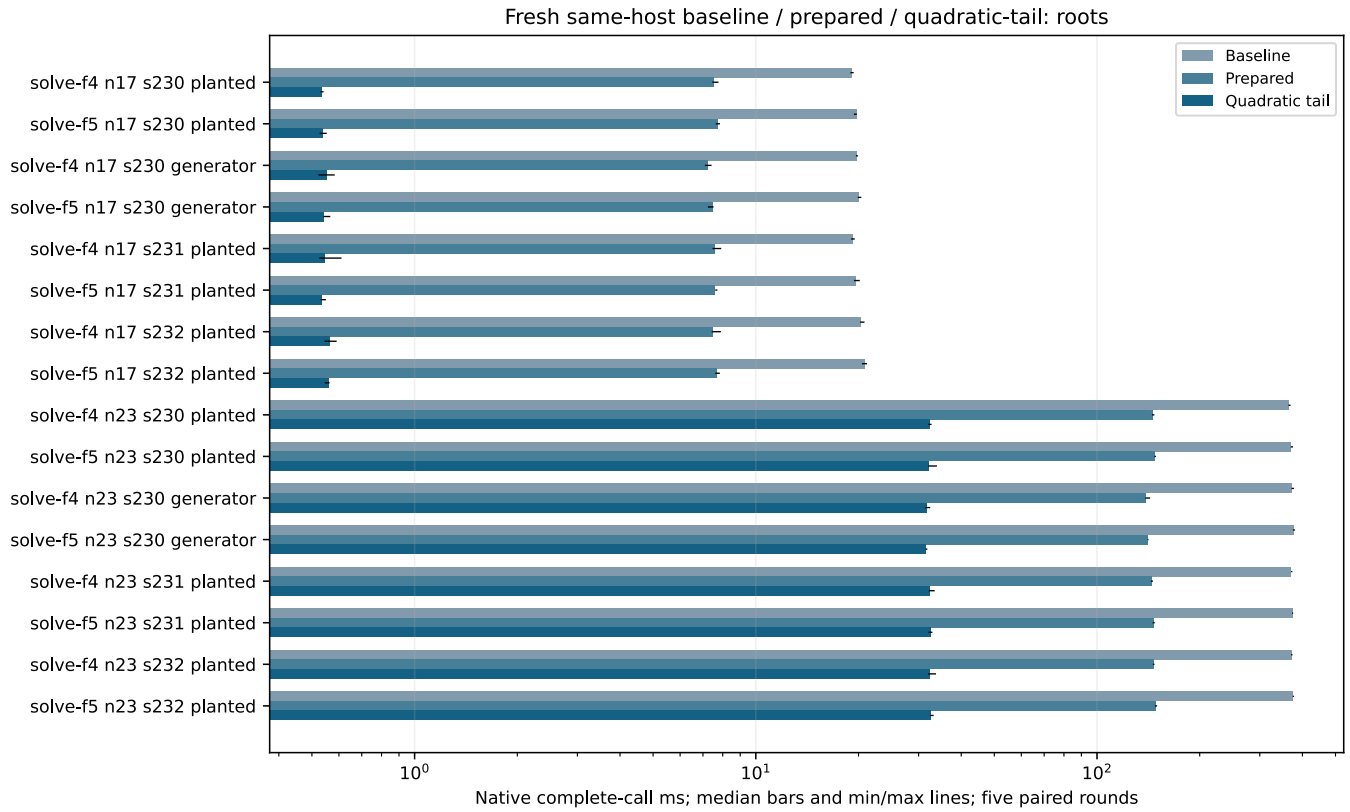
Original n17/n23 fixtures have16/22 Boolean variables on icv1-f2m17-tm101-00378d4e / icv1-f2m23-tm5197-1f85e9e1. Planted seed230 and fixed generator control plus planted holdouts231/232; random=true means generator here. n19's36-variable admission gap remains separate. No fixture or baseline is replaced by the full-DLP standard subspaces.

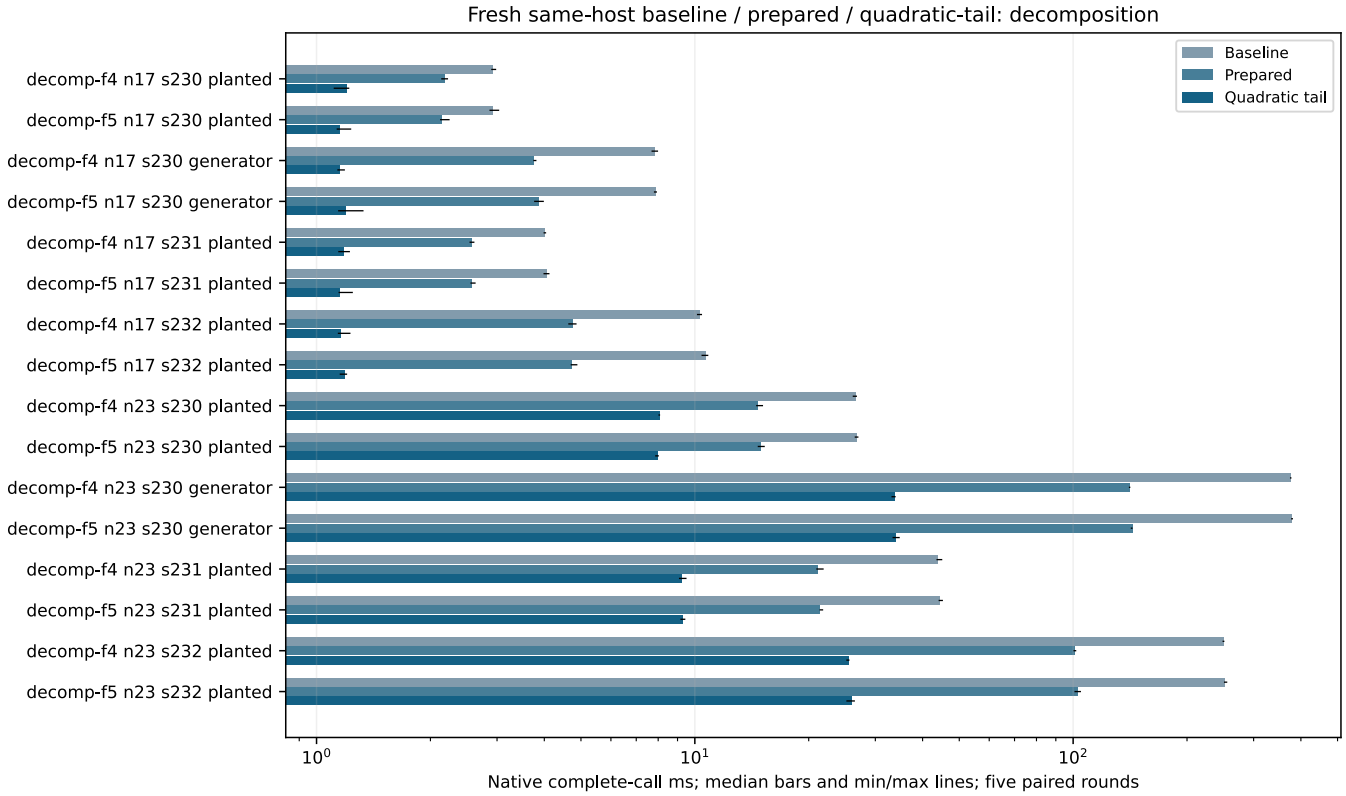
All eighteen arms are rerun with this source's declared policies. Complete original roots and first-witness decomposition remain separate; common curve/base construction is outside component timings. Fresh target ANF, solving, lifting and internal verification are charged to decomposition; Semaev fresh table setup is charged once. Independent original-root/group audits run after elapsed capture.

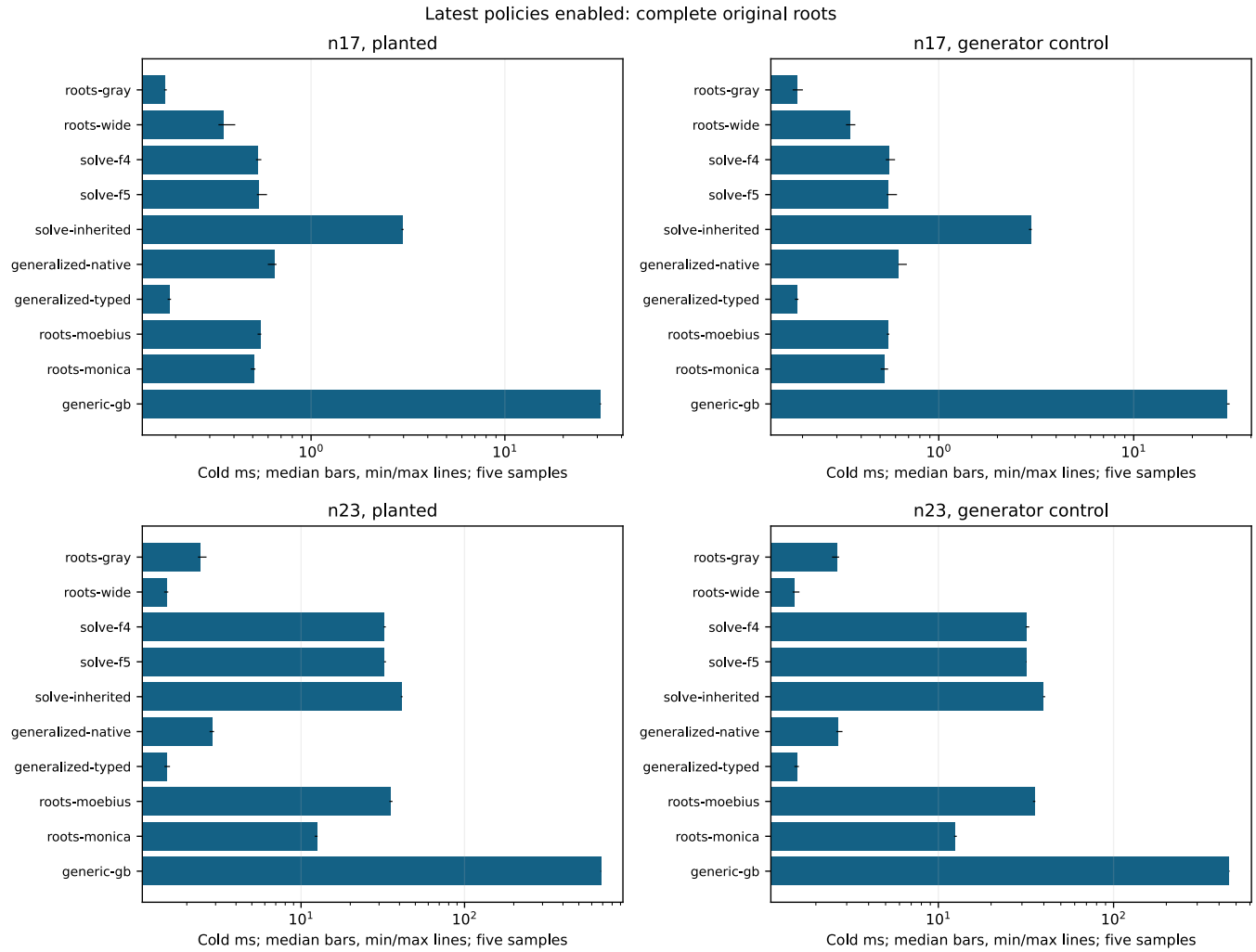
Five paired rounds, first eligible of at most three attempts, exact original outputs, exact3125-resample CI and max(2%, twice A/A noise) gate. Rejected attempts retained: 40. All four multiworker controls neutral: True. Every slower/neutral/missing cell remains in native-summary.json.

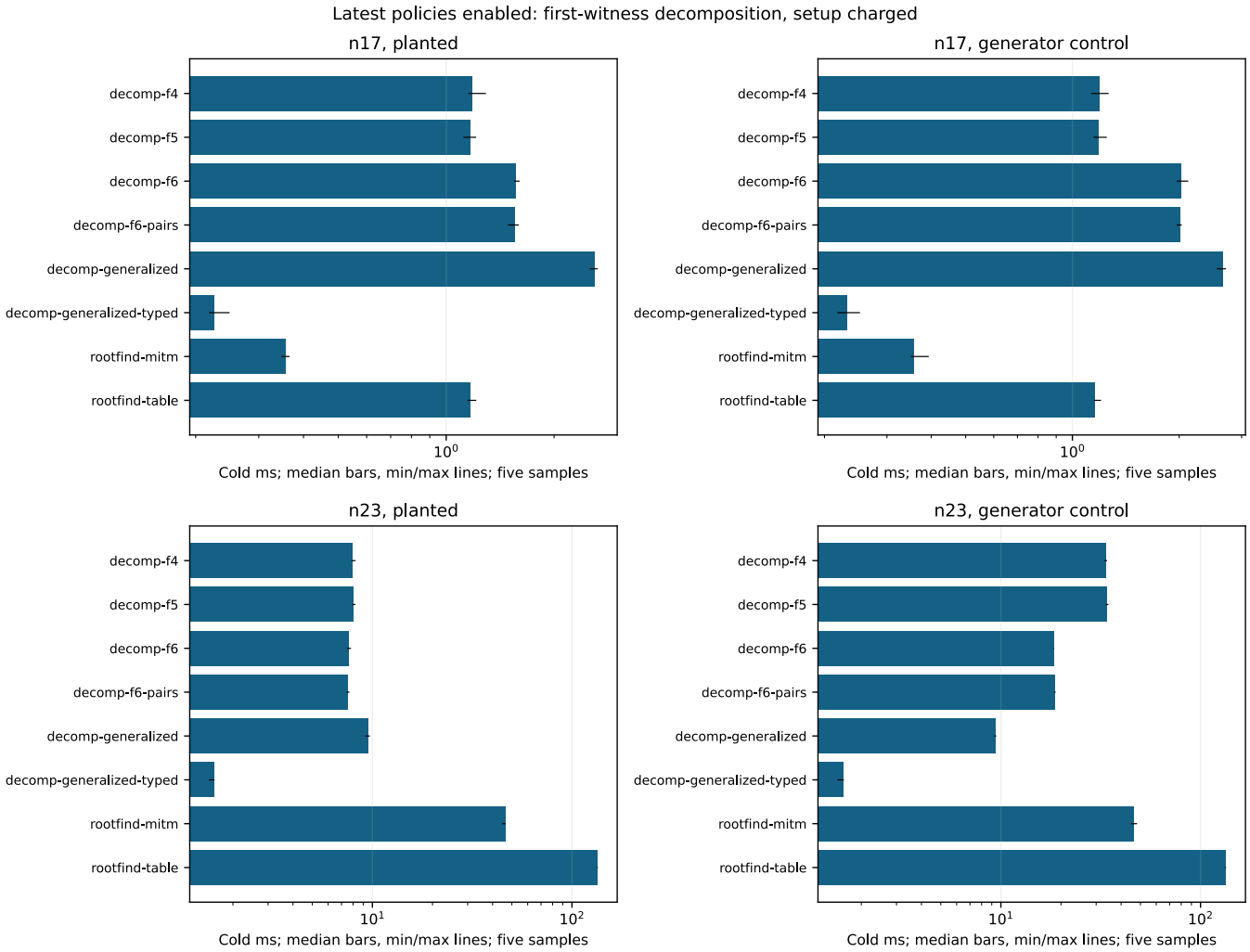
Prepared is a combined active-layout/top-degree/linear-tail/direct-packing ablation. Quadratic-tail retains an algebraic root reduction, then exhaustively completes at most16 free quadratic variables using the existing Gray kernel; original coordinates, root order, caps and verification remain enforced. This is a hybrid engineering change, not a novel enumeration algorithm. Other split rules/custom node oracles keep recursion.

The matrix switches stay opt-in. Source/protocols: 9bed104b0a, PROTOCOL.md and the frozen runner/protocols in the raw tree. Sources0acfc55b7f,71c4c99d85 and9bed104b0a are separate experiments. The final adaptive variants preserve cheap degree-two refutations below16 original variables; legacy policy names remain for replay. Full-table setup uses the same exact sorted tuples serially under an explicit one-worker resource contract. No cross-host pooling, external-fastest, Arm64/GPU, whole-IC/rho or hard-family8x conclusion follows from component calls.




Fresh same-host baseline / prepared / quadratic-tail: decomposition
Baseline
Prepared
Quadratic tail
decomp-f4 n17 s230 planted
decomp-f5 n17 s230 planted
decomp-f4 n17 s230 generator
decomp-f5 n17 s230 generator
decomp-f4 n17 s231 planted
decomp-f5 n17 s231 planted
decomp-f4 n17 s232 planted
decomp-f5 n17 s232 planted
decomp-f4 n23 s230 planted
decomp-f5 n23 s230 planted
decomp-f4 n23 s230 generator
decomp-f5 n23 s230 generator
decomp-f4 n23 s231 planted
decomp-f5 n23 s231 planted
decomp-f4 n23 s232 planted
decomp-f5 n23 s232 planted
10^0
10^1
10^2
Native complete-call ms; median bars and min/max lines; five paired rounds


Latest policies enabled: complete original roots
n17, planted
n17, generator control
n23, planted
n23, generator control
roots-gray
roots-wide
solve-f4
solve-f5
solve-inherited
generalized-native
generalized-typed
roots-moebius
roots-monica
generic-gb
Cold ms; median bars, min/max lines; five samples


Latest policies enabled: first-witness decomposition, setup charged
n17, planted
n17, generator control
n23, planted
n23, generator control
decomp-f4
decomp-f5
decomp-f6
decomp-f6-pairs
decomp-generalized
decomp-generalized-typed
rootfind-mitm
rootfind-table
10^0
10^1
10^2
Cold ms; median bars, min/max lines; five samples

All paired recursive-solver outcomes

Arm	n	Seed	Target	Before ms	Candidate ms	Ratio	Decision
solve-f4	17	230	P	19.1212	0.5369	35.6046	faster
solve-f5	17	230	P	19.7560	0.5390	36.6558	faster
decomp-f4	17	230	P	2.9276	1.1984	2.4902	faster
decomp-f5	17	230	P	2.9228	1.1492	2.5456	faster
solve-f4	17	230	G	19.7640	0.5545	35.6418	faster
solve-f5	17	230	G	20.0131	0.5440	36.7548	faster
decomp-f4	17	230	G	7.8249	1.1523	6.7928	faster
decomp-f5	17	230	G	7.8630	1.1968	6.6347	faster
solve-f4	17	231	P	19.2579	0.5475	35.1766	faster
solve-f5	17	231	P	19.6641	0.5362	36.5913	faster
decomp-f4	17	231	P	4.0124	1.1773	3.3832	faster
decomp-f5	17	231	P	4.0555	1.1543	3.4485	faster
solve-f4	17	232	P	20.2883	0.5631	36.5337	faster
solve-f5	17	232	P	20.8924	0.5612	37.5904	faster
decomp-f4	17	232	P	10.2842	1.1591	9.0028	faster
decomp-f5	17	232	P	10.6986	1.1877	9.0156	faster
solve-f4	23	230	P	365.4568	32.4994	11.2209	faster
solve-f5	23	230	P	371.0485	32.2656	11.5008	faster
decomp-f4	23	230	P	26.6001	8.0531	3.3003	faster
decomp-f5	23	230	P	26.8354	7.9718	3.3690	faster
solve-f4	23	230	G	372.9637	31.8003	11.7343	faster
solve-f5	23	230	G	376.9289	31.6133	11.9231	faster
decomp-f4	23	230	G	376.4585	33.6907	11.2094	faster
decomp-f5	23	230	G	378.6604	33.9865	11.2168	faster
solve-f4	23	231	P	371.1977	32.3034	11.4721	faster
solve-f5	23	231	P	374.1379	32.6527	11.4581	faster
decomp-f4	23	231	P	43.8187	9.2394	4.7426	faster
decomp-f5	23	231	P	44.2896	9.2923	4.8347	faster
solve-f4	23	232	P	372.3018	32.4804	11.4624	faster
solve-f5	23	232	P	376.4863	32.5846	11.5649	faster
decomp-f4	23	232	P	249.6084	25.5159	9.7947	faster
decomp-f5	23	232	P	251.6068	26.0410	9.6692	faster

Latest native complete-call ms (sample count)

Boundary	Method	n17 planted	n17 generator	n23 planted	n23 generator
complete_roots	roots-gray	0.1763 (5)	0.1887 (5)	2.4222 (5)	2.6600 (5)
complete_roots	roots-wide	0.3505 (5)	0.3499 (5)	1.5199 (5)	1.5208 (5)
complete_roots	solve-f4	0.5325 (5)	0.5546 (5)	32.3955 (5)	31.7698 (5)
complete_roots	solve-f5	0.5384 (5)	0.5503 (5)	32.3086 (5)	31.8293 (5)
complete_roots	solve-inherited	2.9710 (5)	2.9921 (5)	41.0426 (5)	39.7848 (5)
complete_roots	generalized-native	0.6481 (5)	0.6235 (5)	2.8599 (5)	2.6983 (5)
complete_roots	generalized-typed	0.1864 (5)	0.1874 (5)	1.5138 (5)	1.5616 (5)
complete_roots	roots-moebius	0.5459 (5)	0.5481 (5)	35.1703 (5)	35.4786 (5)
complete_roots	roots-monica	0.5044 (5)	0.5261 (5)	12.5095 (5)	12.5085 (5)
complete_roots	generic-gb	31.1077 (5)	30.3052 (5)	683.4834 (5)	456.7527 (5)
decomposition	decomp-f4	1.1847 (5)	1.1897 (5)	7.9602 (5)	33.3862 (5)
decomposition	decomp-f5	1.1653 (5)	1.1867 (5)	7.9927 (5)	33.8387 (5)
decomposition	decomp-f6	1.5680 (5)	2.0172 (5)	7.6039 (5)	18.4249 (5)
decomposition	decomp-f6-pairs	1.5511 (5)	2.0109 (5)	7.5319 (5)	18.5602 (5)
decomposition	decomp-generalized	2.5994 (5)	2.6576 (5)	9.4829 (5)	9.3980 (5)
decomposition	decomp-generalized-typed	0.2254 (5)	0.2309 (5)	1.5980 (5)	1.6203 (5)
decomposition	rootfind-mitm	0.3573 (5)	0.3580 (5)	46.2176 (5)	46.0340 (5)
decomposition	rootfind-table	1.1700 (5)	1.1550 (5)	134.0259 (5)	133.0683 (5)

Four-worker matrix controls

Arm	n	Before ms	After ms	Ratio	Decision
matrix-f4	17	8.0974	7.6580	1.0345	neutral
matrix-f5	17	7.9634	8.6035	1.0104	neutral
matrix-f4	23	60.2267	60.3187	1.0115	neutral
matrix-f5	23	58.4961	58.6101	1.0049	neutral

Instrumented matrix phase medians, ms; separate nonadditive observations

Arm	n	Target	Policy	Build ms	Reduce ms	Readback ms	Calls
solve-f4	17	P	baseline	8.8964	9.1975	0.3906	126.0
solve-f4	17	P	prepared	2.6233	4.5930	0.0324	63.0
solve-f4	17	P	quadratic-tail	0.2919	0.1264	0.0003	1.0
solve-f5	17	P	baseline	9.2812	9.3071	0.3891	126.0
solve-f5	17	P	prepared	2.8517	4.5901	0.0289	63.0
solve-f5	17	P	quadratic-tail	0.2962	0.1252	0.0003	1.0
decomp-f4	17	P	baseline	1.1210	0.9685	0.0396	12.0
decomp-f4	17	P	prepared	0.8520	0.5393	0.0022	6.0
decomp-f4	17	P	quadratic-tail	0.2304	0.1263	0.0003	1.0
decomp-f5	17	P	baseline	1.1093	0.9562	0.0395	12.0
decomp-f5	17	P	prepared	0.8517	0.5248	0.0019	6.0
decomp-f5	17	P	quadratic-tail	0.2282	0.1260	0.0002	1.0
solve-f4	17	G	baseline	9.3767	9.4614	0.3842	126.0
solve-f4	17	G	prepared	2.6764	4.2383	0.0269	63.0
solve-f4	17	G	quadratic-tail	0.2933	0.1171	0.0003	1.0
solve-f5	17	G	baseline	9.6866	9.4661	0.3861	126.0
solve-f5	17	G	prepared	2.8719	4.2658	0.0242	63.0
solve-f5	17	G	quadratic-tail	0.2995	0.1180	0.0003	1.0
decomp-f4	17	G	baseline	3.3386	3.3576	0.1371	44.0
decomp-f4	17	G	prepared	1.3378	1.5510	0.0088	22.0
decomp-f4	17	G	quadratic-tail	0.2255	0.1176	0.0003	1.0
decomp-f5	17	G	baseline	3.4257	3.3285	0.1363	44.0
decomp-f5	17	G	prepared	1.4139	1.5459	0.0080	22.0
decomp-f5	17	G	quadratic-tail	0.2353	0.1176	0.0003	1.0
solve-f4	23	P	baseline	159.8091	191.8960	6.0619	1022.0
solve-f4	23	P	prepared	39.0778	103.4899	0.2849	511.0
solve-f4	23	P	quadratic-tail	10.8679	16.8551	0.0063	63.0
solve-f5	23	P	baseline	163.6334	193.0544	6.1095	1022.0
solve-f5	23	P	prepared	40.6718	103.8710	0.2704	511.0
solve-f5	23	P	quadratic-tail	11.0028	16.6557	0.0044	63.0
decomp-f4	23	P	baseline	11.2678	12.3148	0.3824	64.0
decomp-f4	23	P	prepared	4.9917	7.0951	0.0190	32.0
decomp-f4	23	P	quadratic-tail	3.1113	2.3100	0.0014	7.0
decomp-f5	23	P	baseline	11.5059	12.2149	0.3832	64.0
decomp-f5	23	P	prepared	5.1991	7.1437	0.0173	32.0
decomp-f5	23	P	quadratic-tail	3.0834	2.2995	0.0009	7.0
solve-f4	23	G	baseline	165.4343	193.0466	6.0516	1022.0
solve-f4	23	G	prepared	39.9080	96.1717	0.2549	511.0
solve-f4	23	G	quadratic-tail	11.1639	15.9021	0.0059	63.0
solve-f5	23	G	baseline	169.5292	193.0257	6.0629	1022.0
solve-f5	23	G	prepared	41.6756	96.0039	0.2487	511.0
solve-f5	23	G	quadratic-tail	11.3115	15.6300	0.0039	63.0
decomp-f4	23	G	baseline	166.8718	193.8212	6.0476	1022.0
decomp-f4	23	G	prepared	39.4889	95.8989	0.2466	511.0
decomp-f4	23	G	quadratic-tail	10.7112	15.9513	0.0035	63.0
decomp-f5	23	G	baseline	169.9726	192.9853	6.0428	1022.0
decomp-f5	23	G	prepared	41.4014	96.1738	0.2527	511.0
decomp-f5	23	G	quadratic-tail	11.0830	15.9250	0.0038	63.0

Full-table four-worker cold-call controls

Arm	n	Before ms	After ms	Ratio	Decision
rootfind-table	17	1.3923	1.3404	1.0583	neutral
rootfind-table	23	63.2043	63.1086	0.9923	neutral